

Skeptic, Reality, and Randomness: Derandomization in Game-Theoretic Probability

Computer Science Honours Thesis

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Honours Thesis Presentation

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Presentation Overview

- Introduction
- Background: Pseudo-randomness
- Background: Game-Theoretic Probability
- Modified Skeptic–Reality Game
- Main Result: PRGs Keep Capital Bounded
- Proof Idea
- Future Directions

How I Got Into Randomness

- Are coin flips truly random?

How I Got Into Randomness

- Are coin flips truly random? Complexity theory has an answer.
- **Pseudo-randomness:** appearing random to computationally bounded observers.
- **PRG (Pseudo-Random Generator):** expand a small truly random seed into a longer pseudo-random sequence.
- **Derandomization:** replace randomness in an algorithm with deterministic pseudo-random output while keeping behavior “as if random.”

Classical vs Game-Theoretic Probability

- Classical probability: describes randomness via probability measures.
- Game-theoretic probability: models randomness via a sequential game.
- Key players:
 - **Skeptic:** tries to exploit patterns to increase capital (never goes negative),
 - **Reality:** produces outcomes (controls Skeptic's capital),
 - **Forecaster:** optionally announces expectations or prices.

Coin-Tossing Game

Players: Forecaster, Skeptic, Reality

1. Forecaster announces $p_n \in [0, 1]$,
2. Skeptic announces stake $M_n \in \mathbb{R}$,
3. Reality announces $x_n \in \{0, 1\}$,
4. Capital updates: $K_n = K_{n-1} + M_n(x_n - p_n)$.

Collateral duties:

- Skeptic: keep $K_n \geq 0$,
- Reality: prevent $K_n \rightarrow \infty$.

Borel–Cantelli Phenomenon in Coin Tossing

Miyabe–Takemura (2015):

$$\sum_{n=1}^{\infty} p_n < \infty \iff \sum_{n=1}^{\infty} x_n < \infty$$

- Intuition: finitely many predicted heads $\Leftrightarrow$ finitely many actual heads.
- Skeptic can **force** this: if equivalence fails, Skeptic's capital diverges.
- Reality can **comply**: ensure equivalence while keeping capital bounded.
- Key problem: Reality moves after Skeptic, giving a strong "second-player advantage.", thus making their result somewhat trivial

- Goal: study a **simultaneous-style** Skeptic–Reality game where Reality cannot react after seeing Skeptic’s bets.
- Replace true randomness with a **PRG**, asking whether pseudo-random play suffices against computationally bounded Skeptic.
- Investigate: can Reality remain deterministic and keep Skeptic’s capital bounded?

Modified Skeptic–Reality Game

The S-R Game

Players: Skeptic and Reality

- Game played in rounds $i = 1, 2, 3, \dots$
- Round i : Reality outputs block $x^{(i)} = (x_{i1}, \dots, x_{ii}) \in \{0, 1\}^i$
- Skeptic bets sequentially within the block: stake M_{ij} per bit

Payoff for the j -th bet:

$$M_{ij}(x_{i1}, \dots, x_{i,j-1})(2x_{ij} - 1), \quad |M_{ij}| \leq 1$$

Pseudo-Random Generators (PRG)

$$R_i : \{0, 1\}^{\lceil \log i \rceil} \rightarrow \{0, 1\}^i$$

- Seed: $\lceil \log i \rceil$ bits; output: i -bit block
- (i, δ_i) -PRG against circuits of size i^2 if

$$|\Pr[C(R_i(z)) = 1] - \Pr[C(u) = 1]| \leq \delta_i$$

for all size- i^2 circuits C

- Meaning: no small circuit can distinguish output from truly random beyond δ_i

Round Gain and Capital

Round gain:

$$F_i(x^{(i)}) = \sum_{j=1}^i M_{ij}(x_{i1}, \dots, x_{i,j-1})(2x_{ij} - 1) \in [-i, i]$$

Capital after t bets: $K_t, K_0 = 1$

Total bets by round i :

$$T_i = \frac{i(i+1)}{2}, \quad K_{T_i} = 1 + \sum_{k=1}^i F_k(x^{(k)})$$

Goal: choose Reality's blocks so Skeptic's capital stays bounded.

- Skeptic adapts bets to previously revealed bits
- Each stake function

$$M_{ij} : \{0, 1\}^{j-1} \rightarrow [-1, 1]$$

is computable by a Boolean circuit of size $\leq i$

- Interpretation: adaptive but computationally bounded

Main Result: PRGs Keep Capital Bounded

Setup: EXP-computable (i, δ_i) -PRG R_i against size i^2 circuits

Assume: $\sum_{i=1}^{\infty} i\delta_i < \infty$

Then: for every admissible Skeptic strategy S , Reality can pick seeds z_i so that

$$x^{(i)} = R_i(z_i), \quad \sup_i K_{T_i} \leq 1 + 2 \sum_{i=1}^{\infty} i\delta_i < \infty$$

Proof Idea: One Round at a Time

Fix an admissible Skeptic strategy S :

$$F_i(x) = \sum_{j=1}^i M_{ij}(x_1, \dots, x_{j-1})(2x_j - 1) \in [-i, i]$$

- Each F_i computable by circuit of size $O(i^2)$
- Normalize: $H_i(x) = \frac{F_i(x)+i}{2i} \in [0, 1]$
- Threshold tests $C_{i,t}(x)$ computable by size- $O(i^2)$ circuits
- PRG fools such circuits: $|\mathbb{E}_z[H_i(R_i(z))] - \mathbb{E}_u[H_i(u)]| \leq \delta_i$
- Use layer-cake representation(tail-sum formula): integrate threshold tests over $t \in [0, 1]$ to bound $\mathbb{E}[H_i(R_i(z))]$
- Hence, exists seed z_i with $F_i(R_i(z_i)) \leq 2i\delta_i$

Proof Idea: Summing Over Rounds

Choose z_i each round such that $F_i(R_i(z_i)) \leq 2i\delta_i$

$$K_{T_n} = 1 + \sum_{i=1}^n F_i(x^{(i)}) \leq 1 + 2 \sum_{i=1}^n i\delta_i$$

If $\sum_i i\delta_i < \infty$, then

$$\sup_n K_{T_n} \leq 1 + 2 \sum_i i\delta_i < \infty$$

Capital is uniformly bounded at round endpoints.

- Weaken Skeptic admissibility condition?
- Improve circuit-size assumptions?
- More efficient seed search than EXP-time?
- Converse question: bounded capital $\Rightarrow$ PRG existence?
- Extension to sequential game-theoretic probability?

Thank you!

Questions?